

**FLOODSENSE: AN ANDROID-BASED EXPERT SYSTEM FOR FLOOD
SUSCEPTIBILITY ASSESSMENT AND PRE-EVENT PREPAREDNESS IN
BACOR CITY**

Undergraduate Thesis
Submitted to the Faculty of the
Department of Computer Studies
Cavite State University - Bacoor City Campus
City of Bacoor, Cavite

In partial fulfillment of the
requirements for the degree
Bachelor of Science in Computer Science

REYMART V. GOC-ONG
MARTIN LORENZ D. JUANITES
ROMAR T. PULAO
May 2027

FloodSense: An Android-Based Expert System for Flood Susceptibility Assessment and Pre-Event Preparedness in Bacoor City

Reymart V. Goc-ong
Martin Lorenz D. Juanites
Romar T. Pulao

An undergraduate thesis manuscript submitted to the faculty of the Department of Computer Studies, Cavite State University - Bacoor City Campus, City of Bacoor, Cavite, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science with Contribution No. _____. Prepared under the supervision of _____.

INTRODUCTION

Flooding is one of the most frequent and destructive natural hazards in the Philippines. The country's location within the typhoon belt exposes communities to rainfall events that cause inundation, displacement, damage to property, interruption of livelihood, and loss of life. National disaster statistics continue to document the widespread effects of natural hazards on Filipino families and communities (Philippine Statistics Authority [PSA], 2026). Bacoor City, Cavite, is among the localities that regularly experience flooding. Its low-lying areas and connection to the Imus and Bacoor river systems contribute to persistent inundation during periods of heavy rainfall. Flood-mitigation facilities in the Imus River Basin were established specifically to reduce peak floodwater volume in Cavite, including low portions of Bacoor and its tributaries (Japan International Cooperation Agency [JICA], 2021). These conditions establish the importance of accessible, understandable, and location-specific information that residents can consult before a flood event develops.

To assess the information problem, the researchers surveyed 50 residents across 12 barangays in Bacoor City, with respondents drawn largely from historically flood-affected communities. The findings revealed that flood warnings are commonly disseminated through Short Message Service (SMS), social media, and barangay public-address systems, yet these channels do not consistently provide location-specific susceptibility information before flooding occurs. Only 20% of respondents felt well-informed about which areas in the city are flood-prone, meaning that 80% selected Somewhat Aware, Not Sure, or Not Aware at All. Most respondents also rated the timeliness, clarity, and quality of available flood information near the dissatisfied end of the scale. Most critically, 64% reported making a wrong decision, such as taking a flooded road, delaying evacuation, or misjudging flood conditions, because

adequate area-specific information was unavailable. A key informant interview with the Bacoor Disaster Risk Reduction and Management Office (BDRRMO) confirmed that historical flood records and Geographic Information System (GIS)-based exposure information are maintained institutionally but are not presented to residents through an interactive public platform. A supplementary interview with the City Planning and Development Coordinator Office (CPDCO) also established the availability of elevation, flood hazard, land-use, and barangay-boundary information relevant to geographic assessment.

FloodSense addresses this gap through an Android application that connects expert-provided and expert-validated knowledge with the information and preparedness needs of Bacoor City residents. A Rule-Based Expert System organizes approved facts and conditional rules in a knowledge base and applies an inference process to the user's selected rainfall intensity, rainfall duration, and geographic area. Rule-based decision support can present a traceable basis for a result through knowledge-based and human-understandable explanations (Kostopoulos et al., 2024). In FloodSense, the conclusion is an explainable flood susceptibility assessment expressed as Low, Moderate, High, or Very High. The Decision Support System (DSS) then uses the assessment together with the user's situational inputs to present suitable pre-event preparedness guidance. This function supports, but does not make, the user's final decision; decision-support frameworks organize relevant information and structured alternatives for human decision-making (Alabbad et al., 2022).

The Android application presents the assessment through a dynamic GIS-style map that users can pan and zoom. Selectable geographic polygons display susceptibility through a consistent color scheme, while Check My Area allows a user to position a pin and obtain the assessment associated with that selected location. The map also presents verified evacuation-center locations and relevant details. Users select rainfall intensity from Light, Moderate, Heavy, Intense, or Torrential and select a duration of 1, 3, 6, 12, or 24 hours; the Expert System evaluates the supported scenario against approved geographic and historical information. A separate Web Administration System enables authorized administrators to maintain DSS preparedness content, permitted fixed-table values through canonical CSV export and import, map polygons and points, geographic records, evacuation centers, provenance information, and user accounts. Expert rules and the inference algorithm remain protected from ordinary administrator access. Through this arrangement, FloodSense translates institutional knowledge into an accessible susceptibility assessment, spatial visualization, and practical pre-event preparedness support without claiming current-

condition prediction or substituting for official BDRRMO and Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) warnings.

Statement of the Problem

This study aimed to develop an Android application that provides location-specific flood susceptibility assessment and pre-event preparedness support for Bacoor City through a Rule-Based Expert System, a Decision Support System, and a dynamic geographic map. The research questions were grounded in the survey of 50 residents across 12 barangays, key informant interviews with the BDRRMO and CPDCO, and a review of related literature and official information. These sources were used to establish community needs, institutional knowledge sources, geographic requirements, rainfall parameters, preparedness content, and the functional boundaries of the proposed application and its Web Administration System.

In Bacoor City, flooding is a persistent and consequential reality. Survey results reveal that 92% of respondents have personally experienced flooding, with 58% describing the flooding as severe or waist-deep and higher. Despite this exposure, only 20% consider themselves well-informed about flood-prone areas. Mean satisfaction scores for information quality, timeliness, and clarity ranged from 2.62 to 2.72 on a 5-point scale, clustering at the lower end of the Neutral range and tending toward dissatisfaction. These information deficiencies have direct consequences: 64% of respondents reported making wrong decisions due to insufficient flood information, including taking flooded routes, failing to evacuate on time, and misjudging conditions. Open-ended responses further identified late or delayed information at 26% and insufficient or absent location-specific warnings at 20% as the leading concerns. How can an Android application using an Expert System provide residents with understandable, location-specific flood susceptibility assessments based on expert-validated knowledge, rainfall scenarios, and geographic information?

The problem also concerns the accessibility and operational maintenance of institutional knowledge. The BDRRMO maintains historical flood records, exposure information, preparedness protocols, and evacuation-center information, while the CPDCO maintains geographic and planning data. Current dissemination practices, including social media, email, and barangay coordination, do not fully address the public's need for on-demand, situation-specific guidance. A community-facing platform must therefore communicate approved information while ensuring that map objects, preparedness content, permitted fixed-table values, data sources, and administrative accounts can be maintained

by authorized personnel while expert rules and inference logic remain protected. How can a Rule-Based Expert System, dynamic GIS-style map, Decision Support System, and Web Administration System organize this institutional knowledge into an explainable assessment and appropriate pre-event preparedness guidance for Bacoor City residents?

Taken together, these gaps reveal a disconnect between institutional flood knowledge and the decisions residents must make before flooding occurs. The overarching question of the study is: How can FloodSense integrate expert-provided and expert-validated knowledge, rule-based inference, scenario-based rainfall input, interactive geographic visualization, Check My Area, evacuation-center information, and structured decision support to strengthen flood susceptibility awareness and pre-event preparedness in Bacoor City, while remaining a complementary public-information tool that does not perform real-time flood prediction or supersede official warnings?

Objectives of the Study

Generally, the study aimed to develop an Android application that uses a Rule-Based Expert System to assess flood susceptibility and a Decision Support System to provide pre-event preparedness support for residents of Bacoor City through explainable expert knowledge, rainfall scenarios, and dynamic geographic visualization.

Specifically, it aimed to:

1. identify the prevailing challenges, pain points, and needs of Bacoor City residents and relevant local government offices concerning flood information access, timeliness, clarity, location-specific awareness, and pre-event decision-making through a structured survey of 50 respondents across 12 barangays, a key informant interview with the Bacoor Disaster Risk Reduction and Management Office (BDRRMO) to validate survey findings and assess institutional knowledge and data availability, and a supplementary key informant interview with the City Planning and Development Coordinator Office (CPDCO) to identify available geographic, elevation, flood-hazard, land-use, and barangay-boundary information, all conducted during the Empathize phase of the Design Thinking methodology;
2. analyze expert-provided and expert-validated flood knowledge, BDRRMO historical flood records and preparedness protocols, CPDCO geographic information, PAGASA rainfall classifications, evacuation-center information, and user situational requirements to determine the facts, conditional rules,

susceptibility levels, rainfall parameters, geographic entities, and DSS content required by the knowledge base and inference process;

3. develop an Android application using Flutter and Dart, supported by Django, Django REST Framework, GeoDjango, and PostgreSQL/PostGIS, that integrates a Rule-Based Expert System, a protected knowledge base and fixed inference process, PAGASA-aligned rainfall input, a dynamic GIS-style map using flutter_map, selectable color-coded susceptibility polygons, Check My Area, evacuation-center locations, a Decision Support System for pre-event preparedness, user account and onboarding functions, and a separate Web Administration System with a Leaflet-based administrative map for authorized maintenance of DSS content, geographic records, map polygons and points, permitted fixed-table values through canonical CSV export and import, provenance, evacuation centers, and users;
4. test the system's core modules by implementing unit testing, integration testing, system testing, rule and inference validation, application programming interface communication testing, spatial-query testing, database-operation testing, map interaction testing, DSS decision-flow testing, role-based administrative testing, and end-to-end validation to ensure that susceptibility assessment, geographic selection, dynamic visualization, preparedness guidance, and administrative data management operate accurately and consistently;
5. evaluate the system's overall level of user satisfaction in terms of accessibility, reliability, usability, security, maintainability, and portability of the system using ISO/IEC 25010.

Theoretical Framework

The theoretical framework of FLOODSENSE: AN ANDROID-BASED EXPERT SYSTEM FOR FLOOD SUSCEPTIBILITY ASSESSMENT AND PRE-EVENT PREPAREDNESS IN BACOR CITY illustrates how a Rule-Based Expert System, a dynamic Geographic Information System map, and a separate Decision Support System are integrated into an Android and web platform to provide location-specific flood susceptibility information and pre-event preparedness support to residents of Bacor City. Approved institutional data and expert-validated knowledge are managed through the Django Web Administration System and stored in PostgreSQL with PostGIS. The Flutter Android application communicates with the Django backend through secured application

programming interfaces to request assessments, maps, explanations, preparedness guidance, and verified evacuation resources. The framework keeps susceptibility classification, preparedness guidance, spatial visualization, and administrative publishing as coordinated but clearly defined functions.

[Insert the original FloodSense theoretical framework diagram in this location.]

Authentication and Onboarding Module. This module manages resident registration and login using an email address and password. The Django backend verifies credentials, applies password rules, supports email-based password recovery, and issues the session or token required for protected mobile requests. A user's local application state cannot grant administrator privileges because every protected action is checked on the server. During first launch, the onboarding screen explains the purpose of FloodSense, the rainfall controls, map legend, susceptibility meanings, Check My Area procedure, Decision Support System role, data date or version, privacy information, and safety disclaimer. The disclaimer emphasizes that the application provides scenario-based susceptibility and preparedness information rather than a live flood forecast or official warning.

PAGASA-Aligned Rainfall and Location Input Module. This module allows the resident to define the assessment scenario through plain-language rainfall intensity and duration selections. The baseline intensity labels are Light, Moderate, Heavy, Intense, and Torrential, while the supported durations may include 1, 3, 6, 12, and 24 hours. The internal thresholds and wording must follow the final PAGASA reference approved for the study. The resident may select a supported area from the map or manually place a pin through Check My Area. The selected rainfall values and location are planning inputs for the current request; they do not prove current weather conditions or the resident's actual physical location. Unsupported combinations produce a limitation response instead of an assumed classification.

Web Administration and Knowledge Management Module. This module is the controlled maintenance workspace used by authorized administrators. It manages permitted geographic zones and map features, Decision Support System questions and branches, evacuation-center records, public explanations and disclaimers, source and provenance information, user-support actions, and the values used by the fixed assessment table. The Expert System rules, rule conditions, priorities, conflict-resolution procedure, and inference algorithm are protected and are not displayed, created, edited, deleted, or replaced through the ordinary administrator interface. For permitted table-value maintenance, the

administrator exports the canonical comma-separated values (CSV) file generated by FloodSense. The exported file retains the exact table structure required by the algorithm. An authorized editor may change only the values stored within the existing fields; column names, column order, parameter identifiers, required keys, and the table structure must remain unchanged. The administrator may then import the updated CSV file. FloodSense validates the fixed schema before accepting it, and the unchanged algorithm reads the updated values through that same schema. The process does not import expert rules, alter the inference logic, or train a machine-learning model.

Django REST and GeoDjango Processing Module. This module provides the controlled connection between the Android application, administrative workspace, processing services, and database. Django handles server-side authentication, permissions, validation, business logic, and administrative functions. Django REST Framework receives and returns secured JSON and GeoJSON data for the mobile application. GeoDjango provides the spatial models and operations used to store and query polygons and points. When Check My Area submits a coordinate, GeoDjango and PostGIS perform a point-in-polygon query and return the most specific supported zone. Neither the Android application nor the administrator's browser connects directly to the database; all operations pass through the controlled Django services.

Rule-Based Expert System Assessment Module. This module implements a deterministic Rule-Based Expert System. Its knowledge base stores domain knowledge, while its inference procedure applies eligible rules to known facts to reach a conclusion. Knowledge-based and rule-oriented explanations can make decision-support outputs more understandable and traceable (Kostopoulos et al., 2024). FloodSense represents approved knowledge through geographic-zone facts, rainfall categories and durations, decision tables, structured IF-THEN rules, exceptions, priorities, sources, versions, and explanation text. Forward chaining begins with the resolved zone, rainfall scenario, active zone profile, and published rule set; it tests conditions and fires the applicable rules. The primary outputs are Low, Moderate, High, and Very High. Conflicting rules may return Uncertain, missing coverage may return Insufficient Data, and a point beyond approved boundaries returns Outside Supported Area.

Dynamic GIS Map and Check My Area Module. This module presents approved Bacoar City spatial data through a movable map rendered in Flutter using `flutter_map`. Residents can pan, zoom, tap supported zones, open information pop-ups, display map layers, read a text-supported color legend, view evacuation-center markers, and reposition

the Check My Area pin. Supported polygons are colored Low in green, Moderate in yellow, High in orange, or Very High in red based on the current Expert System result. Changing the rainfall scenario causes the same approved polygons to be evaluated and recolored. Map precision is limited to the resolution of the approved boundary and zone data, and administrators cannot assign a public susceptibility class merely by painting a polygon color.

Results Summary and Explainable Result Module. This module presents the assessment in both summary and detailed form. The Results Summary lists supported areas with their current susceptibility badges and may allow searching or filtering by barangay or class. The detailed result identifies the resolved area, parent barangay, rainfall intensity and duration, susceptibility class, plain-language meaning, matched rule or decision-table basis, summarized conditions, source and version information, update date, and any limitation state. The same zone, scenario, and published rule version must produce the same result on the map, in Check My Area, and in the context given to the Decision Support System. A result is marked outdated when its scenario, location, or controlling version changes.

Decision Support System Module. This module provides pre-event preparedness guidance through a deterministic, database-driven decision tree. Decision-support frameworks organize relevant information, scenarios, and structured alternatives to assist human decision-making (Alabbad et al., 2022). FloodSense begins with the latest valid Expert System result as read-only context and combines it with the resident's answers to approved situational questions. The configured path may provide guidance concerning go-bag preparation, family communication, protection of documents and valuables, assistance for vulnerable household members, monitoring of official advisories, and evacuation readiness. The DSS is separate from the Expert System: it does not change the susceptibility class, predict current conditions, or issue an independent evacuation order.

Verified Evacuation Center Resource Module. This module displays only active evacuation-center records that have been verified from BDRRMO or another authorized city source. Each center is stored as a geographic point and may include its name, barangay, address, responsible office, verification status, validation date, and other information approved for public display. PostGIS can rank centers by straight-line geographic distance from the resident's selected pin, and the Android application can open the selected coordinates in an installed navigation application. A nearby ranking or external route does not guarantee that a road is safe or passable, that a center is currently open or has available capacity, or that the suggested route is appropriate during an actual emergency.

User Account and Preferences Module. This module allows authenticated residents to view or update approved profile and convenience information, including their display name, email address, home barangay, optional default rainfall scenario, password actions, privacy information, and logout. Passwords are securely processed by Django and are never shown to administrators in plain text. A manually placed Check My Area coordinate and DSS answers are used to complete the requested operation and are not retained as continuous location or behavior history by default. Resident accounts do not permit editing of Expert System rules, Decision Support System content, official map geometry, rainfall references, or evacuation-center records.

PostgreSQL and PostGIS Database Module. All authoritative relational and spatial records are stored in PostgreSQL with the PostGIS extension. The database contains users and roles, rainfall references, barangays and susceptibility zones, geographic profiles, evacuation-center points, Expert System rules and conditions, Decision Support System nodes and branches, source and review metadata, published versions, and audit history. PostgreSQL enforces relational consistency, while PostGIS supports point-in-polygon and straight-line distance operations. The backend retrieves only the compatible active versions required for a request and returns only public approved data to residents. Internet access is required for fresh synchronization and server-side evaluation; any approved cached information must display its date and version when connectivity is unavailable.

Overall, the framework demonstrates how expert-validated knowledge, structured rule-based inference, geographic processing, explainable results, and a separate preparedness decision tree are combined in one community-accessible system. By converting approved institutional records and expert knowledge into a controlled mobile and web service, FloodSense aims to strengthen location-specific flood susceptibility awareness and support better pre-event preparation without replacing the authority of BDRRMO, PAGASA, barangay officials, or emergency responders.

The diagram presents FloodSense through a top-to-bottom organization. The Internet connection appears above the FloodSense system, the resident and administrator actors appear on the left, approved institutional sources and external display services appear on the right, and the authoritative PostgreSQL/PostGIS database appears below. The stacked modules inside the system boundary show the functions that process user inputs, institutional knowledge, susceptibility rules, spatial data, results, preparedness guidance, evacuation resources, and account information.

System Architecture

[Insert the original FloodSense system architecture diagram in this location.]

The system architecture uses a vertical client-server presentation of the approved FloodSense design. Resident requests enter through authentication, rainfall and location input, and the secured Django service layer. Institutional data and expert review enter through the Web Administration and Knowledge Management Module. The Rule-Based Expert System produces the susceptibility result, which is used by the map and explanation modules and passed as read-only context to the separate DSS. The database stores the controlled relational and spatial records, while basemap and navigation services are used only for display and external directions.

Significance of the Study

The development of FloodSense addresses a critical gap in flood information dissemination by translating expert-provided and expert-validated knowledge, institutional geographic information, and rainfall scenarios into an accessible susceptibility assessment, dynamic map, and pre-event preparedness support. The following stakeholders stand to benefit directly from the study.

For the Bacoor Disaster Risk Reduction and Management Office (BDRRMO). The application provides a complementary channel for communicating approved, location-specific flood susceptibility information and preparedness content beyond social media, email, and barangay coordination. Through the Web Administration System, authorized personnel can maintain DSS guidance, permitted table values through the fixed-schema CSV export-and-import process, map features, geographic information, evacuation-center records, provenance, and user accounts. Expert rules and the inference algorithm remain protected from ordinary administrator access. This supports continuity, accountability, and public accessibility while preserving the authority of official advisories and emergency directives.

For the City Planning and Development Coordinator Office - Bacoor (CPDCO). The study demonstrates a practical community use for geographic and planning information maintained by the office, including elevation, flood-hazard, land-use, and barangay-boundary data. The integration of verified geographic records into spatial polygons, point locations, and area-selection functions illustrates how institutional planning information can support

public understanding of flood susceptibility without presenting FloodSense as a substitute for official technical maps.

For residents of historically flood-affected barangays. The Android application directly responds to the information deficits identified in the survey: 92% of respondents experienced flooding, only 20% felt well-informed about flood-prone areas, and 64% reported wrong decisions associated with insufficient information. Residents can select a rainfall scenario, pan and zoom the dynamic map, inspect color-coded susceptibility polygons, use Check My Area, view verified evacuation-center locations, and receive preparedness guidance suited to the susceptibility result and their stated situation. Plain-language rainfall categories reduce the need to interpret technical meteorological measurements, while the disclaimer directs users to continue following BDRRMO and PAGASA advisories.

For future researchers. The study provides a locally grounded reference for combining knowledge engineering, rule-based inference, decision support, and geographic information in an Android disaster-preparedness application. The documented approach to expert knowledge acquisition, rule validation, spatial data management, dynamic map interaction, administrative maintenance, software testing, and ISO/IEC 25010 evaluation can inform related studies in other flood-prone communities, subject to validation with local experts, datasets, and user populations.

Time and Place of the Study

The research and development of the FloodSense application were conducted using the Design Thinking methodology, beginning with data-gathering activities before the formal study period. The official duration of the study spans from September 2026 to May 2027. Major activities, including requirements analysis, knowledge acquisition, rule structuring, system design, mobile and web development, geographic data preparation, testing, evaluation, and documentation, were carried out at the Department of Computer Studies, Cavite State University - Bacoar City Campus, located in Soldiers Hills IV, Molino, Bacoar City, Cavite. The campus served as the primary venue for software development, academic consultation, integration testing, and preparation of the research manuscript.

Initial data gathering commenced in early May 2026 during the Empathize phase of Design Thinking. A structured survey was administered to 50 residents across 12 barangays in Bacoar City, many of which have historically experienced flooding, from May 5 to May 12, 2026, to capture prevailing challenges in flood information access, timeliness, clarity, and decision-making. On May 12, 2026, a key informant interview was conducted at the Bacoar

Disaster Risk Reduction and Management Office (BDRRMO), located at the City of Bacoar Legislative and Disaster Resilience Building, Bacoar Government Center, Barangay Bayanan, Bacoar City. The interview aimed to validate survey findings, understand the office's workflows and information holdings, and assess the feasibility of a community-facing flood susceptibility and preparedness tool. A supplementary key informant interview was conducted on May 18, 2026 with the Officer-in-Charge of the Bacoar City Planning and Development Coordinator Office (CPDCO), located at the Bacoar Government Center, Barangay Bayanan, Bacoar City. The interview confirmed the availability of elevation maps, flood-hazard maps, land-use maps, and barangay-boundary maps and established the geographic context needed for the study. The outputs of the Empathize phase guided the identification of system requirements and the sources to be subjected to expert validation.

Following data gathering, the study progressed through the Define, Ideate, Prototype, and Test phases within the research timeline. Requirements, interface wireframes, geographic data structures, susceptibility rules, DSS decision flows, mobile screens, and Web Administration System functions were refined through consultations with academic advisers, authorized knowledge sources, and a pilot pool of target users. From March 2027 to April 2027, the study entered the Test and Iteration phases, during which the developed prototype was evaluated by at least 20 Bacoar City residents and representatives or authorized evaluators relevant to the study. Testing focused on rule execution, assessment consistency, map usability, Check My Area, evacuation-center presentation, preparedness guidance, administrative functions, application reliability, and user satisfaction. Feedback collected during this period supported refinements to functionality, clarity, and ease of use before final documentation in May 2027.

Scope and Limitation of the Study

The study developed FloodSense, an Android application using a Rule-Based Expert System for flood susceptibility assessment and pre-event preparedness in Bacoar City. The Expert System applies expert-provided and expert-validated knowledge, approved historical flood information, geographic attributes, and user-selected rainfall parameters through a knowledge base and inference process. Its output assigns an applicable geographic assessment area to one of four susceptibility levels: Low, Moderate, High, or Very High. The Decision Support System then combines the susceptibility result with the user's selected scenario and situational answers to present structured preparedness guidance. FloodSense is designed as a complementary awareness and preparedness platform; its outputs are not

real-time flood guarantees, emergency orders, or substitutes for official BDRMO and PAGASA announcements.

The primary users of the Android application are Bacoor City residents, particularly those from communities that have experienced flooding. Users can register and sign in, review onboarding information and a disclaimer, select a rainfall intensity of Light, Moderate, Heavy, Intense, or Torrential, and select a duration of 1, 3, 6, 12, or 24 hours. The rainfall options represent hypothetical scenarios and are mapped internally to values stored under the fixed fields required by the algorithm. Permitted values may be updated only through the canonical CSV export-and-import process without changing the table structure or inference logic. The application does not require users to enter numerical rainfall measurements, elevation, land use, historical flood frequency, or technical geographic attributes. These facts are maintained in the system through authorized records and are applied by the Expert System during assessment.

The dynamic GIS-style map is limited to the territorial jurisdiction of Bacoor City and to geographic areas for which sufficient expert-validated information and usable spatial records are available. Users can pan and zoom the map, select susceptibility polygons, view the applicable color and assessment details, and keep the same map position while selecting another supported rainfall scenario. The map presents geographic polygons once, while the visual susceptibility style is determined by the Expert System result for the active scenario. Green represents Low, yellow represents Moderate, orange represents High, and red represents Very High. The map is an interactive geographic visualization and does not display live flood extent, water depth, traffic conditions, road closures, river level, evacuation-center occupancy, or satellite rainfall imagery.

Check My Area allows the user to position a pin on the map and request an assessment for that selected point. GeoDjango and PostgreSQL/PostGIS determine which approved geographic polygon contains the point, after which the Expert System evaluates that area using the active rainfall intensity and duration. The result presents the area, rainfall scenario, susceptibility level, and an explanation or safety-status summary. The function is based on the selected pin and does not continuously monitor the user's location. FloodSense does not verify that a specific road is passable or that travel from one point to another is safe under actual conditions.

Evacuation-center locations maintained from authorized records are displayed as map points with available descriptive information. The application may present nearby centers using geographic distance from the selected pin and may open an external

navigation application when the user requests directions. The displayed distance is not proof of the safest route, road accessibility, available capacity, or current operating status unless those details have been verified and recorded by an authorized administrator. Users are instructed to confirm instructions and facility information through official local-government communication channels during an actual emergency.

The Decision Support System is confined to pre-event preparedness. It may ask structured situational questions, such as whether the user is at home, has prepared essential supplies, is responsible for children, older adults, persons with disabilities, or animals, and has identified an evacuation option. The DSS matches the susceptibility result and user inputs with administrator-maintained preparedness content to present reminders, priorities, and available options. It does not issue autonomous evacuation orders, diagnose medical conditions, guarantee personal safety, or make the final decision for the user. A conversational artificial-intelligence assistant is not a core feature of the study; preparedness support is delivered through validated rules, structured questions, and maintained DSS content.

A separate Web Administration System is included for authorized administrators. It provides role-controlled functions for managing DSS questions and guidance, permitted rainfall and geographic values, map polygons and points, geographic attributes, evacuation-center records, provenance and validation information, and user accounts. Expert rules, rule conditions, priorities, conflict-resolution logic, and the inference algorithm are fixed and protected from ordinary administrator access. For permitted value updates, the administrator exports the canonical CSV table, an authorized editor changes only the values within its existing fields, and the administrator imports the file with the same column names, order, identifiers, required keys, and overall structure. Files that do not match the required schema are rejected. The CSV process does not change or import expert rules. A Leaflet-based administrative map supports the maintenance of approved points and polygons, while susceptibility colors presented to residents are produced from Expert System results rather than manual painting of assessment areas. Administrative actions are subject to authentication, authorization, validation, and audit-related information appropriate to the prototype.

The feasible implementation stack consists of Flutter and Dart for the Android application; flutter_map for the resident-facing map; Python Django and Django REST Framework for backend services and application programming interfaces; GeoDjango for geographic operations; PostgreSQL with PostGIS for structured and spatial records; and

Leaflet for the administrative map. The application requires a compatible Android device and an internet connection to retrieve protected knowledge, geographic information, assessment results, preparedness content, accounts, and evacuation-center data. An iOS application and a fully offline mode are outside the study. The system also excludes live sensors, automated meteorological feeds, flood-depth measurement, emergency dispatch, direct control of public warning infrastructure, and continuous background tracking.

The reliability of the susceptibility assessment depends on the completeness, geographic resolution, provenance, and expert validation of the knowledge and data available to the researchers. The study focuses on rainfall-related flood susceptibility within supported Bacoor City areas. Tidal flooding, storm surge, flash flooding originating outside the modeled geographic context, drainage blockages, construction activity, river obstructions, and other rapidly varying conditions are not inferred unless explicitly represented in the approved knowledge base. The four susceptibility classes communicate relative potential under a selected scenario and do not constitute a probabilistic forecast of the time, depth, or exact extent of a flood.

Data privacy and security are addressed through applicable Philippine privacy requirements. The prototype limits collection to information necessary for accounts, administration, testing, and requested application functions. Selected map points and situational answers are processed only to produce the requested assessment and preparedness support according to the implemented design; continuous location history is not part of the study. Access to administrative functions is limited to authorized accounts. The evaluation uses a pilot group and the resulting findings may not be generalized beyond Bacoor City without testing among larger, more diverse populations and validation against the knowledge, geography, and institutional processes of another locality.

Definition of Terms

The following terms are defined operationally as they are used in this study to ensure clarity and consistency throughout the research:

Android Application. The resident-facing FloodSense software installed on a compatible Android device. It provides rainfall scenario selection, susceptibility assessment, the dynamic map, Check My Area, evacuation-center information, DSS guidance, account functions, onboarding, and disclaimers.

Check My Area. A feature that allows a user to position a pin on the dynamic map and obtain the flood susceptibility assessment associated with the approved geographic

polygon containing that selected point.

Color-Coded Susceptibility Levels. The visual scheme used on the dynamic map in which green indicates Low, yellow indicates Moderate, orange indicates High, and red indicates Very High flood susceptibility for the active scenario.

CPDCO (City Planning and Development Coordinator Office). The Bacoor City office that maintains planning and geographic information, including elevation, land-use, flood-hazard, and barangay-boundary materials relevant to the study.

Decision Support System (DSS). The FloodSense component that uses the susceptibility result and the user's situational inputs to present structured, expert-validated pre-event preparedness guidance, reminders, and options while leaving the final decision to the user and official authorities.

Dynamic Map. The GIS-style map in the Android application that supports panning, zooming, geographic selection, color-coded susceptibility polygons, a user-positioned Check My Area pin, and evacuation-center markers.

Expert-Provided and Expert-Validated Knowledge. Flood-related facts, assessment criteria, preparedness guidance, and supporting institutional information supplied or reviewed by authorized subject-matter sources before being encoded in FloodSense.

Expert System. A knowledge-based computer system that applies stored facts and conditional rules through an inference process. In FloodSense, it produces an explainable susceptibility class for a supported geographic area and rainfall scenario.

Flood Susceptibility. The relative potential of an approved geographic area to experience rainfall-related flooding under a supported scenario, expressed in FloodSense as Low, Moderate, High, or Very High. It is not a real-time flood prediction.

Geographic Information System (GIS). A system for organizing, analyzing, and presenting information associated with geographic locations. FloodSense uses GIS functions for polygons, points, containment queries, interactive visualization, and evacuation-center locations.

GeoDjango. The geographic extension of Django used by FloodSense to represent spatial data and perform geographic operations between backend services and the PostGIS database.

Inference Process. The procedure that matches facts from the selected rainfall scenario and geographic area against validated rules in the knowledge base to derive a susceptibility conclusion and supporting explanation.

Knowledge Base. The structured collection of expert-validated facts, protected conditional rules, geographic attributes, fixed table fields, approved values, susceptibility categories, and related provenance used by the Expert System. Ordinary administrators cannot view or modify the protected rules or inference logic.

PAGASA-Aligned Rainfall Input. The user interface for selecting Light, Moderate, Heavy, Intense, or Torrential rainfall and a duration of 1, 3, 6, 12, or 24 hours. These selections represent the rainfall scenarios supported by FloodSense and do not claim to report current weather conditions.

PostGIS. The spatial extension of PostgreSQL used to store and query FloodSense geographic polygons, points, boundaries, selected locations, and evacuation-center coordinates.

Preparedness Provenance. Information identifying the source, validation status, responsible custodian, and relevant date associated with a protected rule, geographic record, preparedness item, permitted table value, or evacuation-center entry.

Rainfall Parameters. The fixed fields used to represent supported rainfall intensity and duration scenarios. Through the CSV workflow, only the values stored within the existing fields may be updated; the field names, order, identifiers, required keys, and structure remain unchanged.

Rule-Based Expert System. An Expert System in which domain knowledge is represented as conditional rules. FloodSense evaluates applicable rules against known scenario and geographic facts to produce a susceptibility assessment.

Web Administration System. The browser-based interface used by authorized administrators to maintain permitted data, DSS content, geographic records, map polygons and points, evacuation centers, provenance information, user accounts, and fixed-table values through canonical CSV export and import. It does not expose or permit changes to expert rules, rule conditions, priorities, conflict-resolution logic, or the inference algorithm.

REVIEW OF RELATED LITERATURE

This chapter presents literature and studies that provide the basis for the flood-susceptibility, geographic-information, decision-support, public-communication, and software-quality requirements of FloodSense. The sources are examined according to their relevance to a scenario-based Android application for Bacoar City. Methods, factors, formulas, and weights from other locations are not treated as FloodSense parameters unless they are supported by local evidence and expert validation.

Related Foreign Literature

Flood Susceptibility and Spatial Factors

Flood susceptibility expresses the relative tendency of an area to experience flooding under defined conditions. Debnath et al. (2024) demonstrated that geospatial and expert-informed approaches can combine multiple flood-conditioning factors and classify areas into ordered susceptibility zones. The study also identified data scarcity, spatial resolution, validation, and the availability of long-term hydrological records as important limitations. These concerns support the use of explicit limitation states in FloodSense when the available Bacoar data cannot justify a classification.

Nguyen et al. (2024) combined a Geographic Information System and an analytical hierarchy process to assess urban flood susceptibility using multiple factors, expert consultation, historical flood points, and receiver operating characteristic validation. The study supports clear factor definitions, documented expert judgment, spatial preparation, and independent validation. Its factors and weights are not transferred to FloodSense because their applicability to Bacoar City has not been established.

Interactive Mapping and Risk Communication

Oubennaceur et al. (2021) found that interactive story maps can combine hazard maps, explanatory text, and user-oriented interaction to communicate flood risk to non-specialists. Clear legends, understandable categories, careful color use, and limited technical language are relevant to the FloodSense map and explanation screens. A map color must be accompanied by text and supporting information so that the user is not required to interpret a technical layer without context.

Lin (2023) examined relationships among flood-risk information seeking, application use, coping beliefs, collective efficacy, and community action. The findings support connecting a susceptibility result to understandable and source-backed preparedness

information. They do not support presenting a software output as an official order or a guarantee of safety.

Decision Support and Explainability

Alabbad et al. (2022) developed a flood-mitigation decision-support framework that combined maps, scenarios, guidance, and decision-tree analysis. The study illustrates how a Decision Support System can organize relevant alternatives and contextual information without making an unexplained final decision for the user. FloodSense applies this principle to pre-event household preparedness and keeps the Decision Support System separate from the susceptibility classification performed by the Expert System.

Kostopoulos et al. (2024) reviewed explainable decision-support approaches, including visual, rule-based, case-based, natural-language, and knowledge-based explanations. Their review emphasizes the importance of domain knowledge and explanations that people can understand. Galanti et al. (2023) likewise showed that explanation quality must be evaluated with intended users. These findings support displaying the basis and limitations of a FloodSense result while protecting the rule set and inference procedure from ordinary administrator modification.

Related Local Literature

Flood Conditions and Institutional Responsibility

The Japan International Cooperation Agency (JICA, 2021) reported the inauguration of flood-mitigation facilities in the Imus River Basin, including structures intended to reduce flooding in low-lying portions of Bacoor and Imus. The project confirms the relevance of basin-level flood management to Bacoor City and shows that a resident-facing application operates alongside government planning, engineering measures, and official emergency functions.

Scenario-Based Hazard Information

GeoRisk Philippines (2026) presents HazardHunterPH as a reference platform for examining hazard information and cautions users that map outputs are not absolute predictions. This boundary supports source labeling, explicit limitations, and the requirement that FloodSense must not treat a basemap or barangay boundary as sufficient evidence for one susceptibility conclusion.

Local Requirements and Data Governance

The Bacoor Disaster Risk Reduction and Management Office interviews identified the need to preserve authoritative records, avoid unsupported barangay-wide generalization, validate evacuation-center information, and maintain preparedness content from authorized

sources (FloodSense Research Team, 2026c; 2026d). The City Planning and Development Coordinator Office interview identified geographic materials relevant to the study but did not establish that every available layer was already suitable for algorithm use (FloodSense Research Team, 2026e). These records support documented provenance, validation, limitations, and controlled maintenance.

Related Foreign Studies

Recent foreign studies demonstrate several approaches relevant to the FloodSense design. Oubennaceur et al. (2021) focused on interactive communication of mapped flood risk; Alabbad et al. (2022) developed a decision-support framework for mitigation alternatives; Galanti et al. (2023) examined understandable explanations in decision-support systems; Debnath et al. (2024) compared geospatial and expert-informed susceptibility models; and Nguyen et al. (2024) developed and validated an urban susceptibility map. Collectively, these studies show that public-facing flood applications require defensible data, a clearly bounded method, validation, understandable maps, and communication appropriate to the system's evidence and authority.

The methods of the reviewed studies cannot be transferred automatically to Bacoar City. Susceptibility research depends on the selected factors, measurements, geographic resolution, weights, and validation observations. FloodSense therefore relies only on locally supported and approved values within its fixed table structure and returns a limitation state when the available knowledge cannot support a conclusion.

Related Local Studies

Johnson et al. (2021) modeled urban expansion and flood exposure across the Philippines using open geospatial data. The study indicates that urban growth can increase the number of people and developed areas exposed to flooding, supporting the need for spatially explicit preparedness information. Its national-scale results cannot, however, be treated as a location-level Bacoar City assessment without local validation.

Nagumo et al. (2022) created a three-dimensional flood-hazard map for a flood-prone Philippine area and emphasized that visualization can help residents understand local flood characteristics. The study supports accessible map presentation but also demonstrates that a visualization must be connected to a documented method and appropriate local data. Neither local study provides a validated set of fixed values for the FloodSense algorithm.

Synthesis

The reviewed sources establish five requirements for FloodSense. First, susceptibility must be tied to defined conditions and supported geographic resolution. Second, an interactive map requires understandable legends, labels, explanations, and limitations. Third, preparedness guidance must remain separate from susceptibility classification and must leave final decisions to residents and authorized institutions. Fourth, rule-based results should be explainable, while the rule set and inference logic remain protected from ordinary administrative modification. Fifth, values obtained from local sources must retain their provenance and be accepted only through a controlled structure that the fixed algorithm can read.

FloodSense addresses a local information gap by connecting a user-selected rainfall scenario and supported Bacoor City location to a deterministic susceptibility assessment and separately sourced pre-event preparedness guidance. Its contribution is the coordinated use of protected rule-based reasoning, geographic presentation, understandable results, controlled data maintenance, and explicit limitations. It does not adopt another study's formula, factor weights, or conclusions as Bacoor City parameters without local evidence and validation.

Table 1. Comparative Synthesis of Selected Related Studies and Systems

SYSTEM OR STUDY	AUTHOR(S)	METHOD	RELEVANCE AND BOUNDARY
Flood Risk StoryMaps	Oubennaceur et al. (2021)	Interactive risk communication	Supports understandable spatial communication; does not provide Bacoor parameters.
Flood Mitigation DSS	Alabbad et al. (2022)	Web decision-support framework	Supports structured guidance while leaving the final decision to people.
Flood Susceptibility Model	Debnath et al. (2024)	Geospatial and expert-informed models	Supports multiple factors and validation; methods require local evidence.
Urban Susceptibility Map	Nguyen et al. (2024)	GIS and analytical hierarchy process	Supports factor documentation and validation; weights are not transferred.
HazardHunterPH	GeoRisk Philippines (2026)	National hazard-reference platform	Supports source and limitation notices; outputs are not absolute predictions.
FloodSense	Present Study	Fixed rule-based scenario assessment	Provides explainable Bacoor-focused assessment using protected rules and controlled values.

METHODOLOGY

This chapter presents the materials and documented activities used in developing FloodSense, the participants involved in completed requirements gathering, and the treatment of research information that remains incomplete. No result from the ongoing survey or final system evaluation is reported until the corresponding responses, sampling information, and analysis have been completed and validated.

Materials

The FloodSense implementation uses Flutter and Dart for the Android application; Python Django and Django REST Framework for backend services; GeoDjango for geographic operations; PostgreSQL with PostGIS for relational and spatial records; flutter_map for the resident-facing map; and Leaflet for the Web Administration System map. Git supports source management, while interface-design and document-processing software support design and research documentation. The final manuscript will identify only the development and testing tools actually used by the researchers.

[The verified specifications of the development computer and Android test device are not yet available; this information will be inserted after confirmation.]

Method

[The software development life-cycle model and its corresponding diagram are not included because the development process and the model that accurately represents it have not yet been confirmed.]

The documented development activities begin with requirements gathering and evidence review. Survey summaries, interview transcripts, approved concept documents, geographic-data records, system specifications, source materials, and implementation evidence are examined to identify supported requirements and unresolved information. A feature that depends on unavailable data or an unverified relationship is not represented as completed system behavior.

Requirements Analysis

The researchers define the intended users, rainfall-scenario inputs, supported geographic areas, susceptibility outputs, preparedness functions, data sources, limitations,

and administrative roles. Resident requirements and institutional information are compared with the implemented system so that a manuscript statement is retained only when it can be supported by a research record or the application.

System Design

The Android application, backend services, spatial database, Rule-Based Expert System, Decision Support System, and Web Administration System are organized as separate but connected components. The fixed inference procedure receives supported scenario and geographic facts, produces a susceptibility result, and passes that result as read-only context to the preparedness component. The administrative interface maintains only the permitted records and values required by its role.

[The system architecture, process, data, and interface diagrams will be inserted in their designated locations after the design documentation is finalized.]

CSV Parameter-Table Exchange

The Web Administration System provides a controlled export-and-import process for values used by the fixed algorithm. The administrator first exports the canonical CSV table produced by FloodSense. The exported table contains the fixed fields and identifiers required by the system. An authorized editor may update only the values inside those existing fields. The editor must not rename, reorder, add, or remove columns; alter parameter identifiers or required keys; or change the structure of the table.

After the value update, the administrator imports the CSV file into FloodSense. The system verifies that the file uses the expected schema before accepting the values. A file with missing, additional, renamed, or reordered required fields is rejected. The imported values are then read by the existing algorithm through the unchanged table structure. This procedure does not expose or import expert rules and does not modify, retrain, or replace the inference algorithm.

Implementation and Verification

Implementation connects the Android interface to the backend and spatial database through validated requests. Verification covers input validation, authentication and permissions, deterministic assessment behavior, geographic lookup, result consistency, map presentation, guidance separation, CSV schema validation, and safe handling of incomplete or unsupported data. The final list of tests and results will be reported only after execution records are complete.

Participants of the Study

The completed requirements-gathering activities include an initial survey of 50 Bacoar City residents from 12 barangays and a separate supplementary survey of 40 respondents from 14 barangays. The two surveys represent different respondent groups and are therefore reported separately. Key informants include representatives of the Bacoar Disaster Risk Reduction and Management Office and the City Planning and Development Coordinator Office whose responsibilities are relevant to flood information, preparedness, institutional records, and geographic data.

[The new survey and final system evaluation are still being conducted. The approved sampling procedure, participant classifications, final number of valid respondents, response rate, and evaluation results will be inserted after completion and validation.]

Statistical Treatment of Data

Frequencies and percentages are used only for the completed descriptive requirements-survey items for which valid response counts are available. Percentage is computed as the frequency of a response divided by the number of valid responses for the item, multiplied by one hundred. The two requirements surveys are analyzed separately and are not combined into a single sample.

[The statistical treatment for the ongoing survey and final system evaluation has not yet been finalized. Scale scoring, verbal interpretations, sample-size computation, reliability analysis, and evaluation tables will be inserted only after the instrument, sampling procedure, completed responses, and analysis plan have been approved and validated.]

REFERENCES

Articles

- GeoRisk Philippines. (2026). HazardHunterPH: Hazard assessment at your fingertips. Retrieved from <https://hazardhunter.georisk.gov.ph/map>
- International Organization for Standardization. (2023). ISO/IEC 25010:2023 systems and software engineering - Systems and software Quality Requirements and Evaluation (SQuaRE) - Product quality model. Retrieved from <https://www.iso.org/standard/78176.html>
- Japan International Cooperation Agency. (2021, October 1). Flood risk protection project in Cavite inaugurated. Retrieved from <https://www.jica.go.jp/english/overseas/philippine/information/press/2021/211001.html>
- Philippine Statistics Authority. (2026). Component 4: Extreme events and disasters. Retrieved from <https://psa.gov.ph/statistics/environment-statistics/highlights/component-4-extreme-events-and-disaster>

Journals

- Alabbad, Y., Yildirim, E., & Demir, I. (2022). Flood mitigation data analytics and decision support framework: Iowa Middle Cedar Watershed case study. *Science of the Total Environment*, 814, 152768. <https://doi.org/10.1016/j.scitotenv.2021.152768>
- Debnath, J., Sahariah, D., Nath, N., Saikia, A., Lahon, D., Islam, M. N., Hashimoto, S., Meraj, G., Kumar, P., Singh, S. K., Kanga, S., & Chand, K. (2024). Modelling on assessment of flood risk susceptibility at the Jia Bharali River basin in Eastern Himalayas by integrating multicollinearity tests and geospatial techniques. *Modeling Earth Systems and Environment*, 10, 2393-2419. <https://doi.org/10.1007/s40808-023-01912-1>
- Galanti, R., Coma-Puig, B., de Leoni, M., Carmona, J., & Navarin, N. (2023). An explainable decision support system for predictive process analytics. *Engineering Applications of Artificial Intelligence*, 120, 105904. <https://doi.org/10.1016/j.engappai.2023.105904>
- Johnson, B. A., Estoque, R. C., Li, X., Kumar, P., Dasgupta, R., Avtar, R., & Magcale-Macandog, D. B. (2021). High-resolution urban change modeling and flood exposure estimation at a national scale using open geospatial data: A case study of the Philippines. *Computers, Environment and Urban Systems*, 90, 101704. <https://doi.org/10.1016/j.compenvurbsys.2021.101704>
- Kostopoulos, G., Davrazos, G., & Kotsiantis, S. (2024). Explainable artificial intelligence-based decision support systems: A recent review. *Electronics*, 13(14), 2842. <https://doi.org/10.3390/electronics13142842>
- Lin, C. A. (2023). Flood risk management via risk communication, cognitive appraisal, collective efficacy, and community action. *Sustainability*, 15(19), 14191. <https://doi.org/10.3390/su151914191>
- Nagumo, N., Ohara, M., Fujikane, M., Inoue, T., Hiramatsu, Y., & Jaranilla-Sanchez, P. A. (2022). Creation of a 3D flood hazard map for a flood-prone area in the Republic of the Philippines and dissemination of the mapping technology. *E-journal GEO*, 17(1), 123-136. <https://doi.org/10.4157/ejgeo.17.123>

Nguyen, H. N., Fukuda, H., & Nguyen, M. N. (2024). Assessment of the susceptibility of urban flooding using GIS with an analytical hierarchy process in Hanoi, Vietnam. *Sustainability*, 16(10), 3934. <https://doi.org/10.3390/su16103934>

Oubennaceur, K., Chokmani, K., El Alem, A., & Gauthier, Y. (2021). Flood risk communication using ArcGIS StoryMaps. *Hydrology*, 8(4), 152. <https://doi.org/10.3390/hydrology8040152>

Research Records

FloodSense Research Team. (2026a). FloodSense first resident requirements survey summary [Unpublished research record]. Cavite State University - Bacoor City Campus.

FloodSense Research Team. (2026b). FloodSense supplementary resident requirements survey analysis [Unpublished research record]. Cavite State University - Bacoor City Campus.

FloodSense Research Team. (2026c). First key informant interview transcript with the Bacoor Disaster Risk Reduction and Management Office [Unpublished research record]. Cavite State University - Bacoor City Campus.

FloodSense Research Team. (2026d). Second key informant interview transcript with the Bacoor Disaster Risk Reduction and Management Office [Unpublished research record]. Cavite State University - Bacoor City Campus.

FloodSense Research Team. (2026e). Key informant interview transcript with the Bacoor City Planning and Development Coordinator Office [Unpublished research record]. Cavite State University - Bacoor City Campus.